

GURU TEGH BAHADUR INSTITUTE OF TECHNOLOGY



ELECTRICAL SCIENCE

LAB MANUAL

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LIST OF EXPERIMENTS

(Electrical Science Lab) ES -159/160

Branch:- CSE /ECE/IT/CSE-AI&ML/CSE-DS

Sl. No.	Experiment
1	Introduction to various Basic Instruments of Electrical Science and Basic Symbols used in Electrical Engineering
2	To verify Superposition Theorem
3	To verify Thevenin Theorem and find out Thevenin's Equivalent circuit using DC Sources
4	To verify Norton's Theorem and find out Norton's Equivalent circuit using DC Sources
5	To verify Maximum Power Transfer Theorem for DC source
6	To study R-L-C series circuit and draw its phasor diagram
7	To perform open circuit test and short circuit test on single phase transformer
8	Starting and reversing of a DC Shunt motor
9	Starting and reversing of an induction motor
10	Connection of Lamp by single way and Two Way Switch

EXPERIMENT 1: Introduction to various Basic Instruments of Electrical Science and Symbols used in Electrical Engg.

AIM: Introduction to various Basic Instruments of Electrical Science and symbols used in Electrical Engineering

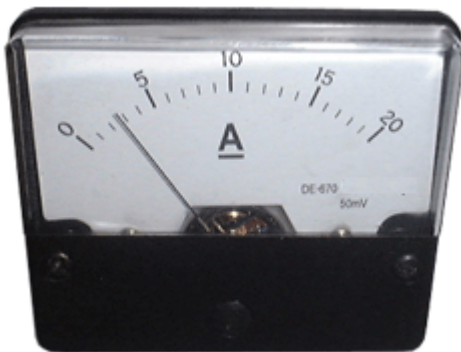
OBJECTIVE: Introduction to various Supply Systems, Ammeter, Voltmeter, Wattmeter, Energy meter, Tachometer, Rheostat, Loading Devices, Transformer.

APPARATUS REQUIRED: Demonstration of various instruments like Ammeter, Voltmeter, Wattmeter, Energy Meter, Tachometer, Rheostat, Various Capacitors, Various Resistors, AC and DC Power Supply.

THEORY OF EXPERIMENT:

AMMETER

Ammeter is employed for measuring of current in a circuit and connected in series in the circuit. As ammeter is connected in series, the voltage drop across ammeter terminals is very low. This requires that the resistance of the ammeter should be as low as possible. The current coil of ammeter has low current carrying capacity whereas the current to be measured may be quite high. So for protecting the equipment a low resistance is connected in parallel to the current coil and it is known as shunt resistance



Analog Ammeter

VOLTMETER

(a) Voltmeter is employed to measure the potential difference across any two points of a circuit. It is connected in the parallel across any element in the circuit. The resistance of voltmeter is kept very high by connecting a high resistance in series of the voltmeter with the current coil of the instrument. The

actual voltage drop across the current coil of the voltmeter is only a fraction of the total voltage applied across the voltmeter which is to be measured.



Analog voltmeter

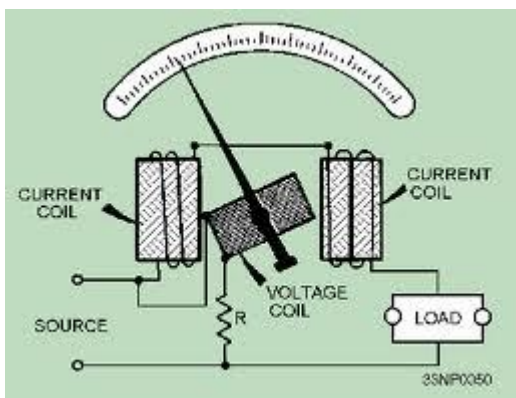
WATTMETER

The measurement of real power in AC circuits is done by using an instrument using Wattmeter. The real power in AC circuits is given by expression

$$VI \cos \Phi$$

where, $\cos \Phi$ is power factor.

A wattmeter has two coils, namely, current coil and pressure coil. The current coil (CC) is connected in series with the load and the pressure coil (PC) is connected across the load. Watt meters are available in dual range for voltages as well as for current



Internal Circuit of Wattmeter

Wattmeter

ENERGYMETER

Energy meter is an instrument which is used to measure the consumption of electric energy in a circuit (DC or AC). It measures energy in kWh. The essential difference between a energy meter and a wattmeter is that the

former is fitted with some type of registration mechanism where by all the instantaneous readings of power are summed over a definite period of time whereas the latter indicates the value at particular instant when it is read.



Energy Meter



TACHOMETER

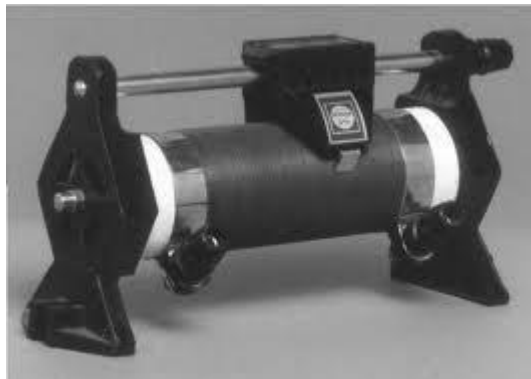
Tachometer is an instrument to measure the speed in (revolutions per minute (r.p.m.)). The speed of a rotating shaft is measured by inserting the tapered projected part of the tachometer into the tapered hole in the rotating shaft speed of which is to be measured.



Tachometer

RHEOSTAT

Rheostats are made up of high resistivity material, like, nickel-chromium iron alloy closely wound over a circular tube. These are available both in single tube and double tube. Inter-turn insulation is provided to avoid short circuiting of turns. The tube of rheostat is made of insulating material, like asbestos. These are employed at places where resistance of a circuit is to be varied without breaking the circuit.



LOADING DEVICES

The most commonly used loading devices are (1) lamp Bank (2) loading Rheostat. Lamp Bank load consists of number of lamps connected to form a load. These are suitably connected and controlled by a no. of switches. The switches are provided in a manner so that it should be possible to switch on any required no. of lamps at a time.

A loading rheostat type of load consists of no. of identical resistive elements. These elements are connected in series or parallel. The rheostat is made up of high resistivity material such as like nickel-chromium. The elements of the load can be designed to take 1A, 2A or 4 A of current.



Loading Rheostat

VARIOUS SUPPLY SYSTEM

(a) **A.C supply systems:** There are two types of supply.

- (i) **Single phase-230V:** In this system we have two wires, one is known as phase/line and the other is neutral. Voltage between them is 230 V.
- (ii) **Three phase - 400 V (line to line):** In his system we have three wires, one for each phase or line. In case the fourth wire is there it is neutral. While voltage between two phases/lines is 400 V, between any phase/line and neutral it is 230 V.

(b) DC Supply System

There are two type of D.C supply system

- (i) From battery: We use rectifiers for 6V or 12V D.C supply current.
- (ii) From generator



DC Supply



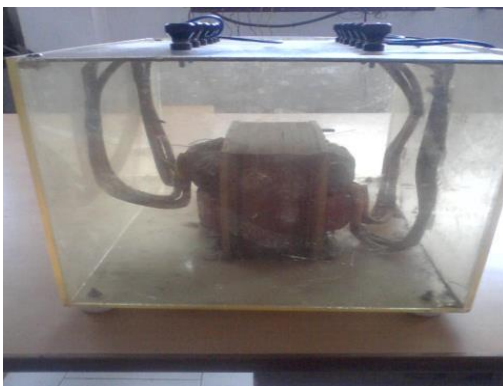
AC Supply

MULTIMETER

Multimeter is a measuring instrument used to measure the current ,voltage and resistance.These can be used to troubleshoot many electrical equipments such as domestic appliances,power supplies etc.



TRANSFORMER: A transformer is a static device which consists of two or more stationary electric circuits interlinked by a common magnetic circuit for the purpose of transferring electrical energy between them. The transfer of electric energy takes place from one circuit to another circuit without change in frequency. Transformer may be for stepping up voltage from low to high or stepping down voltage from high to low.



Single Phase Transformer



Auto Transformer

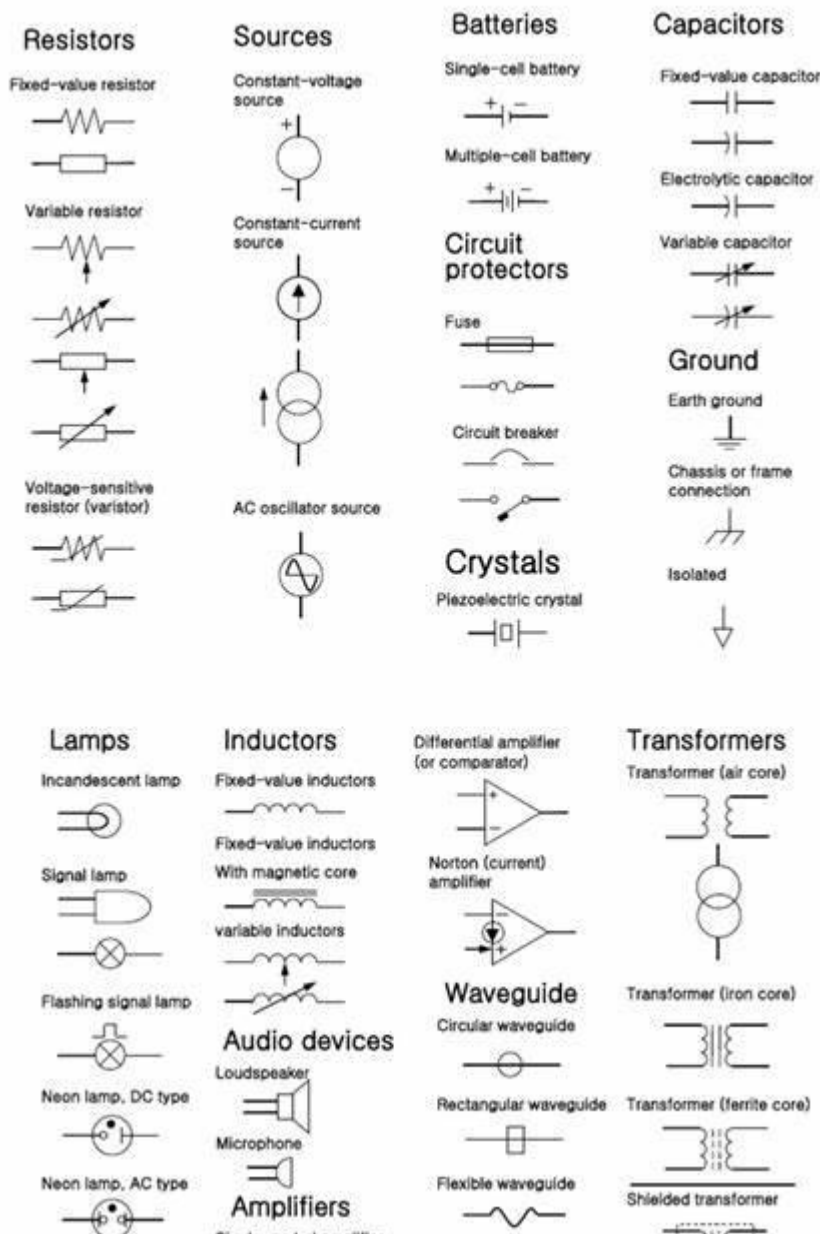
Symbols used in Electrical Engineering:

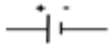
Electrical Symbols are very important in Electrical and Electronics Engineering which virtually represents the actual components in a circuit.

A Electrical Circuit is a graphical representation of the connection of various components represented by their symbols.

Series Circuit: A circuit where bulbs are connected end to end. This type of circuit is known as series circuit. Here the current can flow only in one direction and the same amount of electric current flows through all the bulbs.

Parallel Circuit: Here each bulb is connected to the battery terminals by separate wires. This type of circuit is known as parallel circuit. In this circuit, different amount of current passes through the bulbs.



	Cell
	Battery
	AC source
	Ammeter
	Voltmeter
	Galvanometer
	Resistance
	Variable Resistance
	Bulb/Load
	Closed Switch Key
	Open Switch Key
	Closed Plug Key
	Closed tapping key

REFERENCES

Books

- 1."Electrical Science" by J. B. Gupta
2. "A Text book of Electrical Technology" by B. L. Thereja Vol-11
- 3."Electrical Engineering Fundamentals" by Del Toro
- 4."Electric Circuits" by James Nelson (Pearson publication)
- 5."Basic Electrical Engg." By DC Kulshreshtha, TMHill.

URL's

1. www.brighthub.com
2. www.allaboutcircuits.com
3. www.howstuffworks.com
4. www.nptel.iitm.ac.in

LAB TUTORIALS

1. What are the basic measuring instruments for measuring electrical quantities?
2. What is the working principle of wattmeter and an energy meter?

3. What are the various safety measures to be taken while performing practical work in electrical science lab?
4. Discuss various types of resistors and capacitors?
- 5 Define the term ideal current and ideal voltage source?

EXPERIMENT 2: Verification of Superposition Theorem

AIM: To verify Superposition Theorem.

OBJECTIVE: To apply the principle of Superposition Theorem for electrical network containing independent DC sources.

APPARATUS: Digital multi-meter, power supply, resistance (wire wound), Connecting Wires

THEORY OF EXPERIMENT:

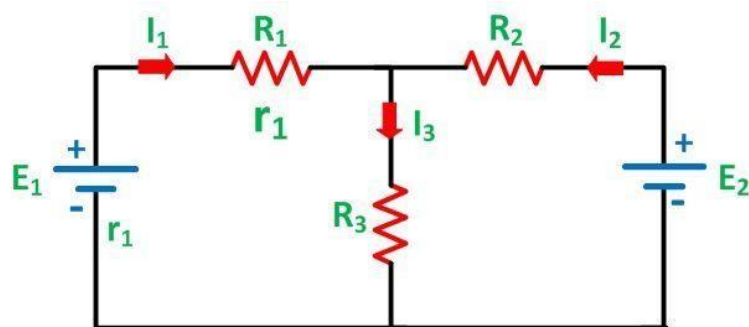
Superposition theorem states that in a linear network containing several independent sources, the overall response at any point in the network equals the sum of responses due to each independent source considered separately with all other independently sources made inoperative(short circuited). To make a source inoperative, it is short circuited leaving behind its internal resistance if it is a voltage source, and it is open circuited leaving behind its internal resistance if it is a current source.

In most electrical circuit analysis problems, a circuit is energized by a single independent energy source. In such cases, it is quite easy to find the response (i.e., current, voltage, power) in a particular branch of the circuit using simple network reduction techniques (i.e., series parallel combination, star delta transformation, etc.).

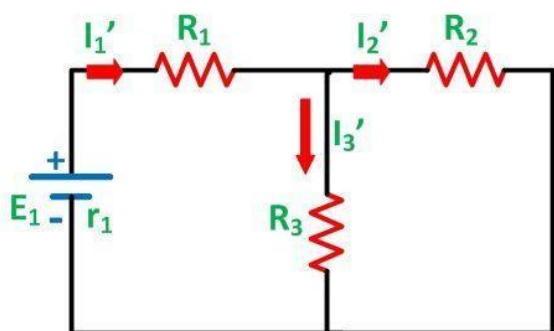
However, in the presence of more than one independent source in the circuit, the response cannot be determined by direct application of network reduction techniques. In such a situation, the principle of superposition may be applied to a linear network, to find the resultant response due to all the sources acting simultaneously.

The superposition theorem is based on the principle of superposition. The principle of Superposition states that the response (a desired current or the voltage) at any point in the linear network having more than one independent source can be obtained as the sum of responses caused by the separate independent sources acting alone. The validity of principle of superposition means that the presence of one excitation sources does not affect the response due to other excitations.

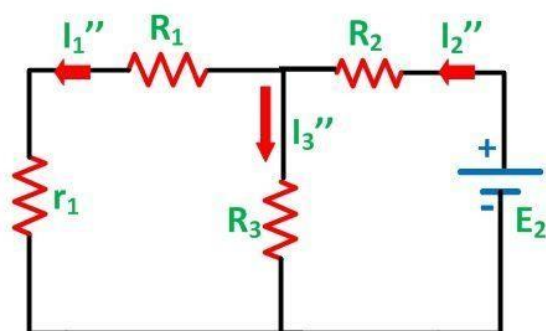
Circuit Diagram:--



Circuit Diagram A



Circuit Diagram B



Circuit Diagram C

Circuit Globe

PROCEDURE:

1. Connect the DC power supply to resistance R1. Adjust voltage of supply to 10V.
2. Connect another DC supply to resistance R2. Adjust voltage to 5 V.
3. Connect the DC ammeter(mA) to resistance R3.
4. Now remove the left hand side of supply and measure and record the current through R3.

$$I_3 = \text{___ ma}$$

5. Remove another supply and measure and record the current through R3.

$$I_3' = \text{___ ma.}$$

6. Now apply both the supplies and measure the current in R3 i.e. I_3'' .

$$\text{Now } I_3'' = I_3 + I_3'$$

OBSERVATION TABLE:

Calculated Values

I_3	I_3'	I_3''

Observed Values

$V_1=10V$	$V_2=5V$	$V_1=10V$
$V_2=0V$	$V_1=0V$	$V_2=5V$
I_3	I_3'	I_3''

CALCULATIONS:

$$I_3 \text{ " (Observed) } = I_3 + I_3'$$

$$I_3 \text{ " (Calculated) } = ? \text{ (by solving using KVL)}$$

$$\% \text{ Error} = (\text{Observed Value} - \text{Calculated Value}) / \text{Calculated Value}$$

RESULT:

The percentage error is found to be __%.

DISCUSSION:

The % error is found to be in the range within 10%. The percentage error is due to observational errors, tolerance errors, calibration of instruments, etc. However, superposition theorem cannot be applied to non-linear network and network containing only dependent sources.

CONCLUSION:

The superposition theorem is verified.

REFERENCES:

Books:

1. Fundamentals of Electrical engineering by Ashfaq Husain.
2. A Textbook of Electrical Technology by B.L Thereja.
3. Electrical Science by J. B. Gupta
4. "Basic Electrical Engg." By DC Kulshreshtha, TMHill.

URLS:

1. www.brighthub.com
2. www.allaboutcircuits.com
3. www.howstuffworks.com
4. www.nptel.iitm.ac.in

LAB TUTORIALS

1. What are the advantages and disadvantages of using superposition Theorem?
2. Why superposition theorem is not applied to non-linear circuits?

3. Can superposition theorem be applied to circuit having A.C sources? If yes, then what will be its requirements?
4. How can superposition theorem be applied to network containing both independent and dependent sources?

EXPERIMENT 3: Verify Thevenin Theorem and find out Thevenin's Equivalent circuit using DC Sources

AIM: To verify Thevenin Theorem and find out Thevenin's Equivalent circuit using DC Sources.

OBJECTIVE: To apply the principle of Thevenin Theorem for electrical network containing independent DC sources and to find out Thevenin's Equivalent circuit.

APPARATUS: Digital Multi-meter, Power Supply, Resistance (wire wound), Connecting Wires

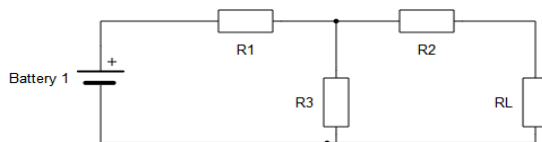
THEORY OF EXPERIMENT:

Sometimes, we wish to determine the response in a single load resistance in a network. Thevenin Theorem enables us to replace the remainder of the network by a simple equivalent circuit. Determining response in the load resistance, then becomes easier. The use of Thevenin Theorem is specially very helpful and time saving when we wish to find the response for different values of load resistance. Thevenin Theorem states that current through a load resistance connected across any two points of an active network can be obtained by the formula:

$$I_L = V_{th} / (R_{th} + R_L)$$

Where V_{th} is the open circuit voltage at the terminals of R_L when R_L is disconnected and R_{th} is the equivalent resistance viewed from the output terminals when all the sources replaced by their internal resistance only (short circuit the voltage sources).

Circuit Diagram:



PROCEDURE:

1. Connect the DC power supply to resistance R1. Adjust voltage of supply to 10V.
2. Connect another DC supply to resistance R2. Adjust voltage to 5 V.

3. Now remove the resistance R_L and measure V_{th} (open circuit voltage) by setting meter in range 0 to 20V .
4. Remove both the sources . Set your meter to measure the resistance in hundreds of ohms range.
Calculate R_{th}
5. Now the Thevenin's equivalent circuit is as in figure.

OBSERVATION TABLE:

Calculated Values

V_{th}	R_{th}	I_L

Observed Values

V_{th}	R_{th}	I_L

CALCULATIONS:

$$I_L = V_{th} / (R_{th} + R_L)$$

Where V_{th} is the open circuit voltage at the terminals when disconnected, R_{th} is the equivalent resistance viewed from the output terminals when all the sources are replaced by their internal resistance. I_L is the current through resistance R_L .

$$\text{Percentage Error} = [(\text{Observed} - \text{Calculated}) / \text{Calculated}] * 100$$

The percentage error is to be found for the current flowing in the load resistance.

RESULT:

The percentage error is found to be __%.

DISCUSSION:

The % error is found to be in the range within 10%. The percentage error is due to observational errors, tolerance errors, calibration of instruments. Moreover, it can be seen that Thevenin Theorem can't be applied to network containing only dependent sources.

CONCLUSION:

The Thevenin Theorem is verified and the Thevenin's equivalent circuit is obtained.

REFERENCES:

.Books

- 1."Electrical Science" by J. B. Gupta
2. "A Text book of Electrical Technology" by B. L. Thereja Vol-11
- 3."Electrical Engineering Fundamentals" by Del Toro
4. "A Text book of Electrical Technology" by B. L. Thereja Vol-1
- 5."Basic Electrical Engg." By DC Kulshreshtha, TMHill.

URL's

1. www.brighthub.com
2. www.allaboutcircuits.com
3. www.howstuffworks.com

LAB TUTORIALS

1. What are the advantages and disadvantages of using Thevenin Theorem?
2. Why Thevenin Theorem is not applicable to non-linear circuits?
3. Can Thevenin Theorem be applied to circuit having A.C sources? If yes, then what is the difference in using DC sources?

EXPERIMENT NO. 4 To verify Norton's Theorem and find out Norton's Equivalent circuit using DC Sources

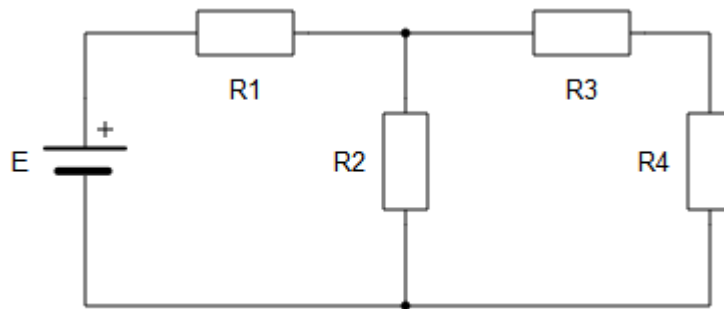
AIM: To verify Norton's theorem.

APPARATUS: DC network kit and connecting wires.

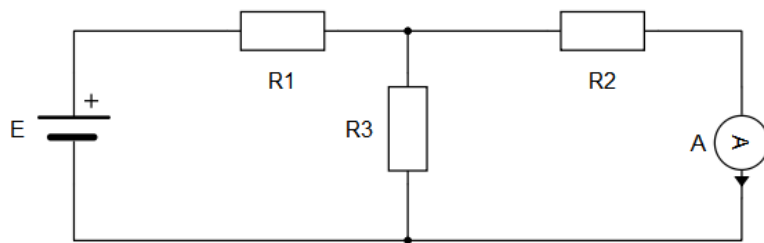
THEORY: Norton's theorem replaces the electrical network by an equivalent constant current source and a parallel resistance. Norton's equivalent resistance $R_N = R_1 * R_2 / R_1 + R_2$ Actual load current in the circuit I_{L1} Theoretical load current $I_{L2} = I_{SC} * R_N / (R_N + R_L)$, I_{SC} is the short circuit current.

CIRCUIT DIAGRAM:

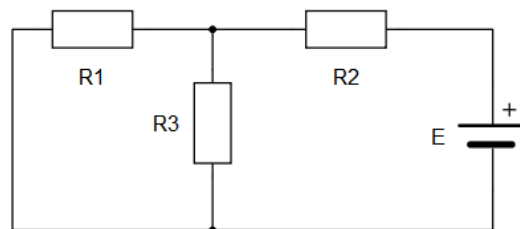
Find current in R_L



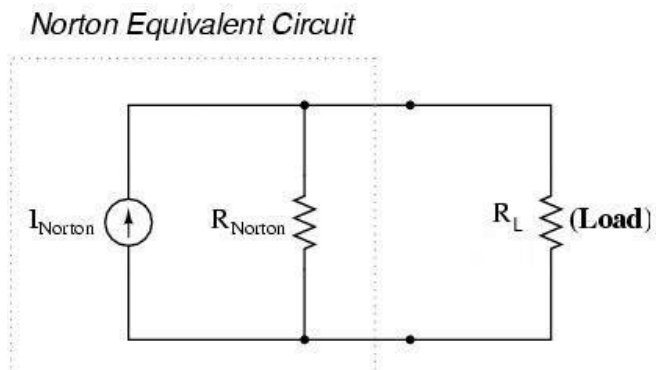
STEP I Short circuit R_L and Find current in AB



STEP II Find R_N



Step III Draw equivalent circuit and Find current in R_L



OBSERVATION TABLE:

SR.NO.	APPLIED VOLTAGE (volts)	I_N (mA)	R_N (Ω)	I_{L1} (mA)	I_{L2} (mA)	<u>ERROR</u> $I_{L1} - I_{L2}$	<u>RESULT</u>

PROCEDURE :

1. Connect the circuit as per the circuit diagram
2. Remove the load resistance
3. Find the Norton's resistance R_N
4. Measure the Norton's current I_N
5. Now measure the current in the load resistance directly
6. Find out the current in the load
7. Using formula find out the current in the load resistance
8. Verify that these two are equal.

RESULT: Norton's theorem is verified

CONCLUSION: In Norton's equivalent circuit the Norton's current source is in parallel with NORTON'S resistance and the load resistance.

Viva QUESTIONS:

Q1 To what type of network Norton's theorem applicable?

Q2 How is R_N connected to I_N ?

Q3 What is placed in place of voltage sources while calculating the R_N ?

Q4 Give an example of unilateral ckt?

Q5. What is unilateral ckt?

Q6. Give one example of the bilateral n/w?

Q7. What is the limitation of Ohm's law?

Q8 What is the reason that ground pin are made of greater diameter in the plugs?

Q9 Where is the voltage divider rule applicable?

Q10 Where is the current divider rule applicable

EXPERIMENT 5: To verify Maximum Power Transfer Theorem for DC source

AIM: To verify Maximum Power Transfer Theorem for a DC source.

OBJECTIVE: To understand the concept of Maximum Power Transfer Theorem to the network consisting of a fixed internal resistance R_S and a variable load resistance R_L and if maximum power is drawn by the network then it is to be proved that $R_L = R_S$.

APPARATUS: one fixed resistor, one variable resistor, digital multi-meter, DC Power supply, connecting wires.

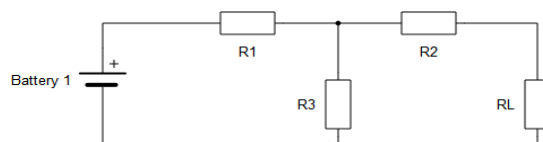
THEORY OF EXPERIMENT: This theorem is applicable for analyzing communication networks. According to this theorem "A resistive load will draw the maximum power from a network when the load resistance is equal to the resistance of the network as viewed from its output terminals, with all energy sources removed leaving behind their internal resistances." If R_L is the load resistance connected across terminals a and b which consist of variable DC supply and internal resistance is R_S , then according to this theorem, the load resistance will draw maximum power when it is equal to R_S i.e. $R_L = R_S$.

And the maximum power drawn = $V_{oc}^2 / 4 R_L$

Where, V_{oc} is the open circuit voltage at the terminals from which R_L is disconnected.

The variable resistor taken should be larger than fixed resistor. Then only power can be calculated.

Circuit Diagram:



PROCEDURE:

1. Connect the circuit as shown in Figure.

2. Measure the value of load current (I_L) for different (suitable) values of load resistance and record them in observation table. By suitable value means that the value of R_L should be equal to, more than and less than R_S .
3. Repeat step 2 for different value of supply voltage (V_S).
4. Also note down the reading in voltmeter connected across R_L . Let it be V .
5. Calculate the value of power accordingly. You will notice that the power goes on increasing as R_L is increasing and after few observations it goes on decreasing.
6. Note down the maximum power and at that point calculate the value of R_L . This R_L should be equal to R_S .
7. Draw a Graph Between P_L and R_L .

OBSERVATION TABLE

S.No	Supply Voltage(V_S)	I_L (Load Current)	V(Voltage)	$R_L=V/I_L$	$P=I_L^2 R_L$

CALCULATION

$$R_L = V/I_L$$

$$P = I_L^2 R_L$$

Note the maximum power point and hence determine the resistance R_L . This R_L is comparable with R_S . These come out to be same.

RESULT:

The maximum power transfer theorem is verified as $R_L = R_S$

DISCUSSION

It can be analyzed that at maximum power, the value of load resistance is equal to internal resistance.

CONCLUSION

The value of load resistance comes out to be almost equal to the internal resistance of the network.

REFERENCES

1. Fundamentals of Electrical engineering by Ashfaq Husain.

2. A Textbook of Electrical Technology by B.L Thereja.

3. "Basic Electrical Engg." By DC Kulshreshtha, TMHill.

URLS:

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2. www.electronics-tutorials.ws/dccircuits
- 3 www.openbookproject.net
4. www.mhhe.com
5. www.opamp-electronics.com
6. www.electronicsteacher.com

LAB TUTORIALS

1. Define power transfer efficiency in Maximum Power Transfer Theorem?
2. What are the practical applications of Maximum Power Transfer Theorem?
3. Can Maximum Power Transfer Theorem be applied to A.C sources?
4. What is the expression of η when $R_L = R_S$?

EXPERIMENT 6: To study R-L-C series circuit and draw its phasor diagram

AIM: To study R-L-C series circuit and draw its phasor diagram.

OBJECTIVE: To study the AC R-L-C series circuit and analyze the phase relationship between voltage and current.

APPARATUS: Ammeter (0-1A), Resistor, Inductor and Capacitor, Variac, Connecting Wires and Digital Multi-meter.

THEORY OF EXPERIMENT: Consider an AC circuit containing resistance of R ohms, inductance of L henries and capacitance of C farads connected in series, as shown in circuit diagram. Let the current flowing through the circuit be of I ampere and supply frequency be f Hz.

- (a) Voltage drop across resistance $V_r = IR$ in phase with I.
- (b) Voltage drop across inductance $V_l = \omega L$ leading I by $\frac{\pi}{2}$ radians or 90° .
- (c) Voltage drop across capacitance, $V_c = I / \omega C$ lagging behind I by $\frac{\pi}{2}$ radian or 90° .

V_l and V_c are 180° out of phase with each other (or reverse in phase), therefore, when combined by parallelogram they cancel each other. The circuit can either be effectively inductive or capacitive depending upon which voltage drop (V_l or V_c) is predominant. Let us consider the case when V_l is greater than V_c .

Phase angle ϕ between voltage and current is given by:-

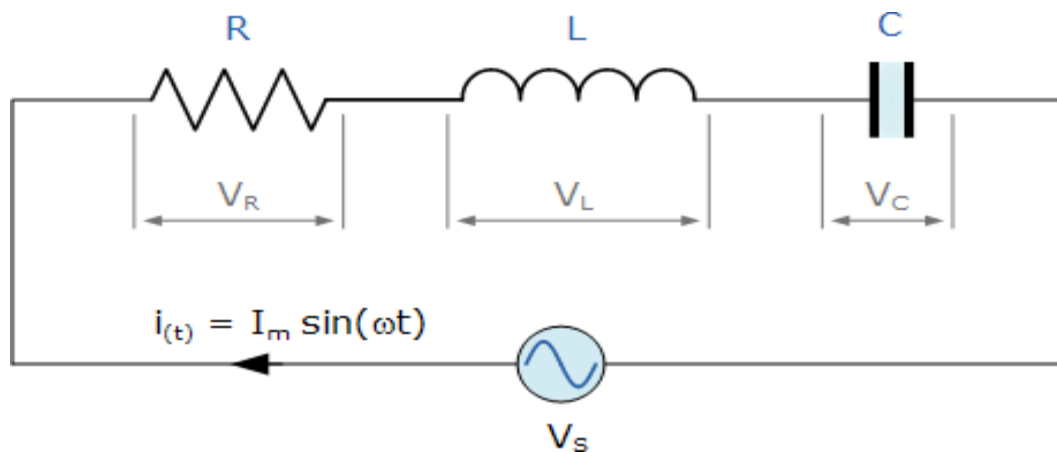
$$\phi = \tan^{-1} V_l - V_c / V_r.$$

Power factor of the circuit is given by:-

$$\cos \phi = R/Z = R/\sqrt{R^2 + (X_l - X_c)^2}$$

Power consumed in the circuit, $P = I^2 R$ or $VI \cos \phi$.

CIRCUIT DIAGRAM:--



PROCEDURE:

1. Connect the circuit as shown in the circuit diagram.
2. Measure the voltage supplied by variac (V_1).
3. Measure the circuit current (I) using AC ammeter.
4. Note down the voltage drop across R, L and C by digital multi-meter.
5. Change the input voltage and then again note down the voltage drop (at least 5 readings)
6. Draw the phasor diagram between V_1 , V_C , and V_R and calculate the phase.

OBSERVATION TABLE

V_i	V_r	V_L	V_C	I_A	R	X_L	X_C	$Z = \sqrt{R^2 + (X_L - X_C)^2}$	$\cos \phi$

CALCULATIONS

1. $R = V_r / I$
2. $X_L = V_L / I$
3. $X_C = V_C / I$
4. $\cos \phi = R / Z$
5. $Z = \sqrt{R^2 + (X_L - X_C)^2}$
6. Draw a phasor diagram showing voltage between V_r , V_L and V_C and calculate its phase angle.

RESULT:

The value of Z is and the value of Φ (phase angle) and power factor is

DISCUSSION

The RLC series circuit has been analyzed and it can be predicted that the circuit has leading power factor. The main precaution to be taken while performing the experiment is that the value of resistance should be small otherwise resonance will take place.

REFERENCES

1. Fundamentals of Electrical engineering by Ashfaq Husain.
2. A Textbook of Electrical Technology by B. L. Thereja.
3. Electrical Science by J. B. Gupta
4. "Basic Electrical Engg." By DC Kulshreshtha, TMHill.

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1. www.nptel.iitm.ac.in
2. www.electronics-tutorials.ws/dccircuits
3. www.openbookproject.net
4. www.mhhe.com
5. www.opamp-electronics.com

LAB TUTORIALS

1. What do you understand by the term power factor in reference to AC circuits ?.
2. What is the importance of power factor?
3. What do you mean by a lagging power factor?
4. What do you mean by a leading power factor?

EXPERIMENT 7: To perform open circuit test and short circuit test on single phase transformer

AIM: To perform open circuit test and short circuit test on a single phase transformer.

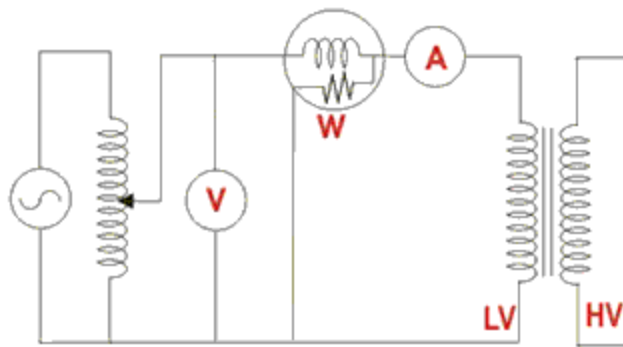
OBJECTIVE: To perform open circuit and short circuit tests and calculate the equivalent circuit parameters Cu losses and Iron losses.

APPARATUS: Single phase Transformer (2kVA), AC Ammeter (0-10 A) AC voltmeter (0-300 V), Wattmeter (10A, 300V), Connecting leads , Variac(230/270V, 20 A).

THEORY OF EXPERIMENT:

The performance of a transformer can be calculated on the basis of its equivalent circuit which contains four main parameters, the equivalent resistance R_{01} as referred to primary (or secondary R_{02}), the equivalent leakage reactance X_{01} as referred to primary. These constants or parameters can be easily determined by two tests, i.e. open circuit test and short circuit test. These are very economical and convenient, because they furnish the required information without actually loading the transformer.

The purpose of the **Open Ckt. Test** is to determine no load loss or core loss and no load I_0 which is helpful in finding X_0 and R_0 . One winding of the transformer usually high voltage winding is left open and the other is connected to its supply of normal voltage and frequency. A wattmeter (W), Voltmeter (V) and ammeter (A) are connected in the low voltage winding, i.e., primary winding in the present case. With normal voltage applied to the primary, normal flux will be setup in the core, hence normal iron losses will occur which are recorded by the wattmeter. As in the primary no load current I_0 is small, Cu loss is negligibly small in primary and nil in secondary. Hence, the wattmeter reading represents practically the core loss under no load condition.

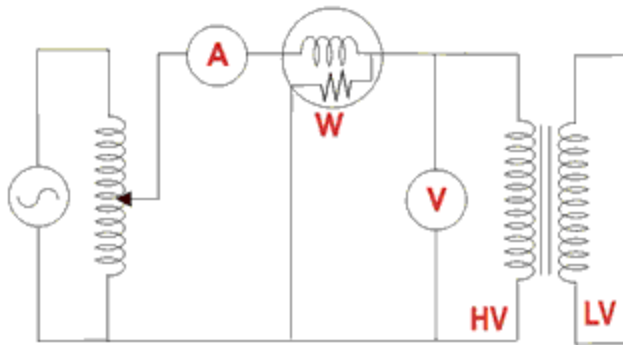


Open Circuit Test on Transformer

For **short circuit test**, one winding usually the low voltage winding, is short-circuited by a thick conductor (or through an ammeter which may serve the additional purpose of indicating rated load current).

A low voltage (usually 5 to 10% of normal primary voltage) at correct frequency is applied to the primary and is gradually and cautiously increased till full-load current is flowing both in primary and secondary (as indicated by the respective ammeters).

Since, in this test, the applied voltage is a small percentage of the normal voltage, the mutual flux ϕ produced is also a small percentage of its normal value. Hence, core losses are very small with the result that the wattmeter reading represents the full load Cu loss or $i^2 R$ loss for the whole transformer, i.e. sum of both primary and secondary Cu losses.. The equivalent impedance of the transformer under short-circuit condition, if V_{sc} is the voltage required to circulate rated load currents, is then given by $Z_{01} = V_{sc}/I_1$.



Short Circuit Test on Transformer

PROCEDURE:

OPEN CIRCUIT:-

1. Switch on the supply, increase the supply in an orderly manner (rated voltage) to 2kVA of transformer. This test is performed on low voltage winding and high voltage winding is open.
2. Note down the readings of wattmeter, voltmeter and ammeter.
3. Compute the value of $\cos \phi$, R_0 and X_0 .

SHORT CIRCUIT:-

1. For short circuit test short the secondary winding. This test is performed on high voltage winding, low voltage winding is short circuited.
2. Increase the voltage applied slowly so that the current flowing in the transformer winding equals the rated value.
3. Record the readings of ammeter, voltmeter and wattmeter; which correspond to short-circuit current, corresponding applied voltage and power with full load current flowing under short circuit condition respectively.
4. Compute the value of short circuit impedance and resistance.

OBSERVATION TABLE

For Open- Circuit test:-

V_1	I_0	W	$\cos \phi$	X_0	R_0

CALCULATIONS:-

$$W = V_1 I_0 \cos \phi$$

$$\text{Therefore, } X_0 = V_1 / I_u, R_0 = V_1 / I_w$$

$$\text{Where } I_w = I_0 \cos \phi, I_u = I_0 \sin \phi$$

OBSERVATIONS:-

For Short Circuit test:-

V	I	W	Z_{01}	X_{01}	R_{01}

CALCULATION

For Short Circuit Test:-

$$W = I^2 R_{01}$$

$$R_{01} = W / I^2$$

$$Z_{01} = V / I$$

$$X_{01}^2 = (Z_{01}^2 - R_{01}^2)$$

RESULT:

The Cu loss is obtained toW and iron loss isW.

DISCUSSION: The open circuit and Short circuit test have been performed and various parameters like R_0 , X_0 , Z_{01} and X_{01} are calculated and losses of transformer i.e. iron and copper loss can be determined.

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1. Fundamentals of Electrical engineering by Ashfaq Husain.
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LAB TUTORIALS

1. Why indirect testing of large size transformers is necessary?
2. When a transformer is energized what types of losses occur in the magnetic frame of the transformer?
3. What do you understand by all day efficiency of a transformer?
4. What type of losses are ignored in the short circuit test on a transformer?

EXPERIMENT NO.8 Starting and speed reversing of a DC shunt motor.

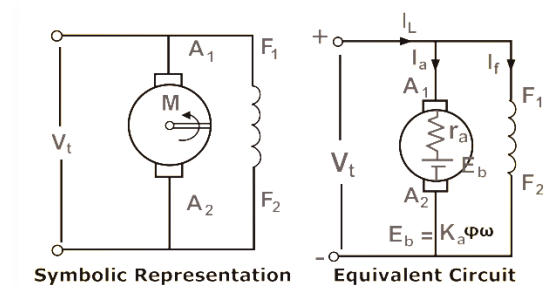
Aim: To Perform Starting and speed reversing of a DC shunt motor.

Objectives:

1. To become familiar with the starting and reversing speed of a DC shunt motor.
2. To become familiar with the starting of DC shunt motor using 3-point starter.

Equipments/ Instruments Required: DC shunt motor, DC Ammeter 0-2A, DC Ammeter 0-20V, DC voltmeter 0-300V, Rheostat 1000 ohm, 1.2A, 3-point starter

Basic theory:



The symbolic representation of a DC Shunt Motor is depicted in the figure above, where F_1 and F_2 are the field terminals, and A_1 and A_2 are the armature terminals of the D.C. Shunt Motor M . The equivalent circuit between the armature terminals A_1 & A_2 is shown. The back emf (E_b) varies with speed ω .

$$E_b = K_a \Phi \omega$$

During starting E_b is zero as $\omega=0$. Therefore starting armature current $I_{ast} = V_t / R_a$ is very high. But the machine winding is designed considering sufficient armature back emf present, which gives armature current much lower than this value. Armature Current $I_a = (V_t - E_b)/R_a$. Hence, the armature winding may not withstand this I_{ast} . To limit the starting current, the resistance is inserted in the armature circuit. It is removed gradually to avoid unwanted loss while running. This resistance is swept to field circuit to reduce the field current thereby to achieve a good speed. Speed is inversely proportional to field current.

$$\begin{aligned} E_b &= V_t - I_a r_a \\ \text{i.e. } K_a \Phi \omega &= V_t - I_a r_a \\ \text{or } \omega &= \frac{V_t - I_a r_a}{K_a \Phi} \end{aligned}$$

Thus, ω is inversely proportional to Φ . The above starting arrangement is implemented through DC Motor Starter.

THREE- POINT STARTER

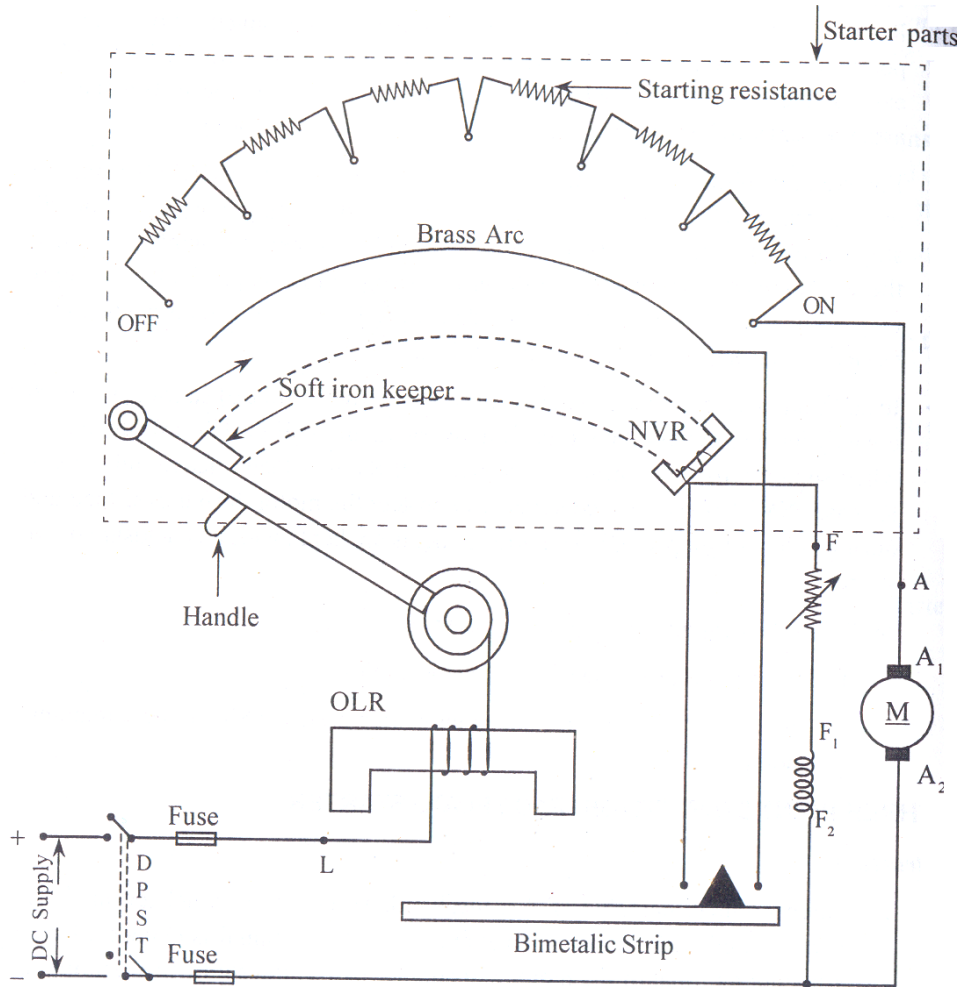
The internal wiring for such a starter is shown in the fig. 1 below. The three terminals of the starting box are marked as A, B, C. One line is directly connected to one armature terminal and one field terminal which are tied together. The other line is connected to point A which is further connected to the starting arm L, through the over –current (or over load) release M.

To start the motor, the main switch is first closed and then the starting arm is slowly move to the right. As soon as the arm makes contact with stud no.1, the field circuit is directly connected across the line and at the same time full starting resistance R_S is placed in series with the armature. The starting current drawn by the armature= $V/(R_A+R_S)$ where R_S is the starting resistance. As the arm is further moved, the starting resistance is gradually cut out till, when the arm reaches the running position, the resistance is all cut out. The arm moved over the various studs against a strong spring which tends to restore it to OFF position. There is a soft iron piece S attached and held by an electromagnet E energized by the shunt current. It is variously known as “HOLD-ON” coil, LOW-VOLTAGE (or NO-VOLTAGE) realize.

It will be seen that as the arm is moved from stud 1 to the last stud, the field current has to travel back through that portion of the starting resistance that has been cut out of the armature circuit. This results in slight decrease of shunt current. But as the value of starting resistance is very small as compared to shunt field resistance, this slight decrease in I is negligible. This defect can, however, be remedied by using a brass arc which is connected to stud no.1.

The normal function of HOLD-ON coil is to hold the arm in the full running position when the motor is in running position. But, in case of failure or disconnecting of the supply or break in the field circuit, it is de-energized thereby releasing the arm which is pulled back by the spring to the OFF position. This prevents the stationary armature from being put across the lines again when the supply is restored after temporary shutdown. This would have happened if the arm were left in the full null position. One great advantage of connecting the HOLD-ON coil in series with the shunt field is that, should the field circuit become open, the starting arm immediately springs back to the OFF position thereby preventing the motor from running away.

The over-current release consists of electromagnet connected in the supply line. If the motor becomes over-loaded beyond a certain predetermined value, then D is lifted and short-circuits the electromagnet. Hence, the arm is released and returns to OFF position.



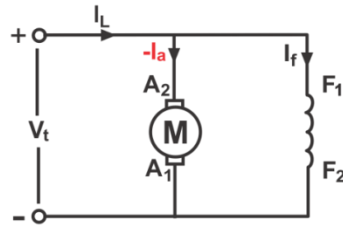
Circuit Diagram

Reversing the speed of DC Motor

Speed of DC shunt motor can be reversed either by interchanging the armature terminals or by interchanging the field terminals.

1. Interchanging the armature terminals (A_1 and A_2)

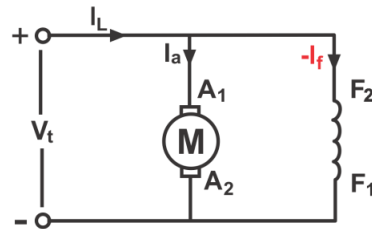
The electromagnetic torque $T_e = K_a \phi I$. If the armature terminals are changed, the direction of armature current will be reversed i.e. the sign I_a will be changed. Therefore the electromagnetic torque, $T_e = K_a \phi (-I_a) = -K_a \phi I_a$. Torque is negative, that means the direction of rotation will change.



2. Interchanging the field terminals (F_1 and F_2)

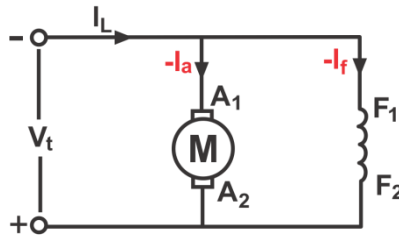
If the field terminals are changed, the sign I_f and therefore sign of ϕ will be changed. Thus the electromagnetic torque

$T_e = K_a(-\phi)I_a = -K_a\phi I_a$. That means the again direction of rotation will change.



3. Interchanging the supply terminals ($L1$ and $L2$)

If the supply terminals are changed then sign of both I_a and I_f will change. Thus, $T_e = K_a(-\phi)(-I_a) = K_a\phi I_a$. That means the torque remains positive and the direction of rotation remains unchanged.



Procedure:

1. The circuit is connected as per the circuit diagram shown.
2. The DC supply is switched on with minimum resistance in the field the field circuit and the starting handle is moved to first position slowly.
3. As soon as the arm makes contact with the first stud of the 3 point starter, the motor starts running.
4. The armature current increases rapidly and settles to a steady value.
5. The armature current, armature voltage and field current at the instant of starting are noted.
6. Gradually all resistances are cut by further moving arm to the right slowly.
7. When the arm reaches the run position, all the resistances are bypassed and starting arm would be attached with electromagnet.
8. The armature current, armature voltage and field current and speed are noted when values settle to a steady value.
9. The direction of rotation of motor is observed at different connection mentioned in table below.
10. The supply is switched off and starting arm comes to off position automatically.

Experimental data:**Table 1**

Obs. No.	Operation mode	Field current (I_f) A	Armature current (I_a) A	Speed (N) rpm
1.	At the start			
2.	Final			

Table 2

Particulars of observation	Speed (N) rpm	Direction of rotation
Connection as per circuit diagram		
Reverse field winding only		
Reverse armature winding only		
Reverse both armature and field winding		

Precautions:

1. Voltage across armature should not be applied without applying voltage to the field winding of a DC motor
2. All connections are to be tightly connected.

Conclusion:

Starting (with 3-point starter) and speed reversal of a DC shunt motor is studied.

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1. Fundamentals of Electrical engineering by Ashfaq Husain.
2. A Textbook of Electrical Technology by B. L Thereja.
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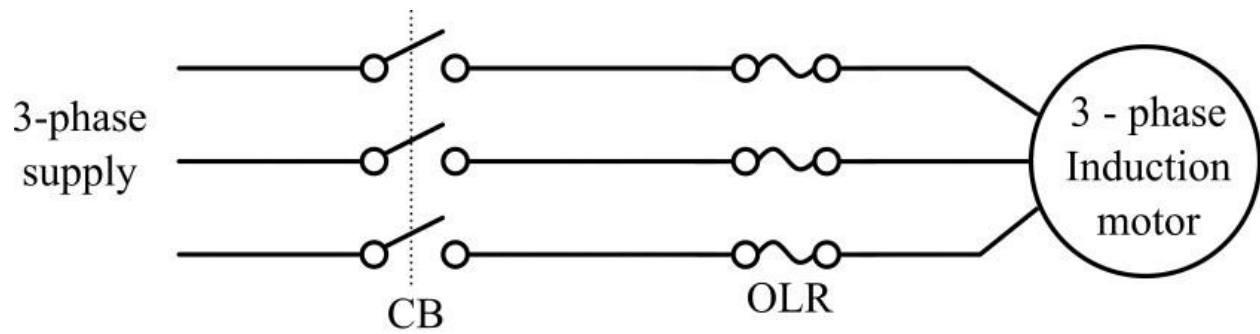
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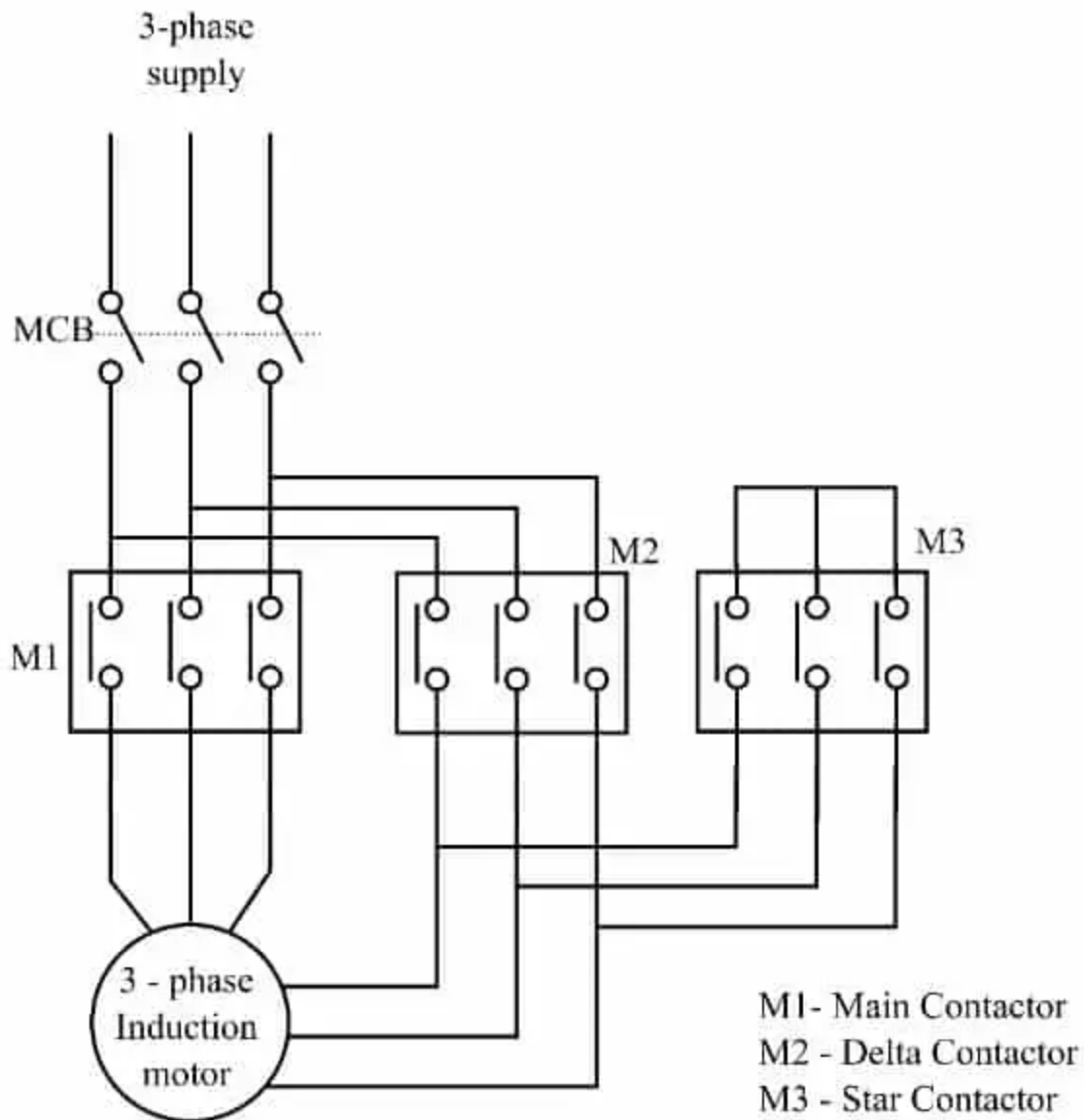
Experiment No 9. Study the starting and Reversing the direction of rotation of a three phase induction motor

AIM: Study the starting and Reversing the direction of rotation of a three phase induction motor
When supply is connected to a three phase induction motor a rotating magnetic field will be set up in the stator, this will link and cut the rotor bars which in turn will induce rotor currents and create a rotor field which will interact with the stator field and produce rotation. Of course this means that the three phase induction motor is entirely capable of self starting. The need for a starter therefore is not, conversely enough, to provide starting but to reduce heavy starting currents and provide overload and no-voltage protection. There are a number of different types of starter including The Direct On-line Starter, Star- Delta Starter ,Auto-Transformer and Rotor resistance.

Direct-on-Line Starter (DOL) The DOL starter switches the supply directly on to the contacts of the motor. As the starting current of an induction motor can be 6-8 times the running current the DOL starter is typically only used for motors with a rating of less than 5kW.



Direct On-Line (D.O.L.) Starter

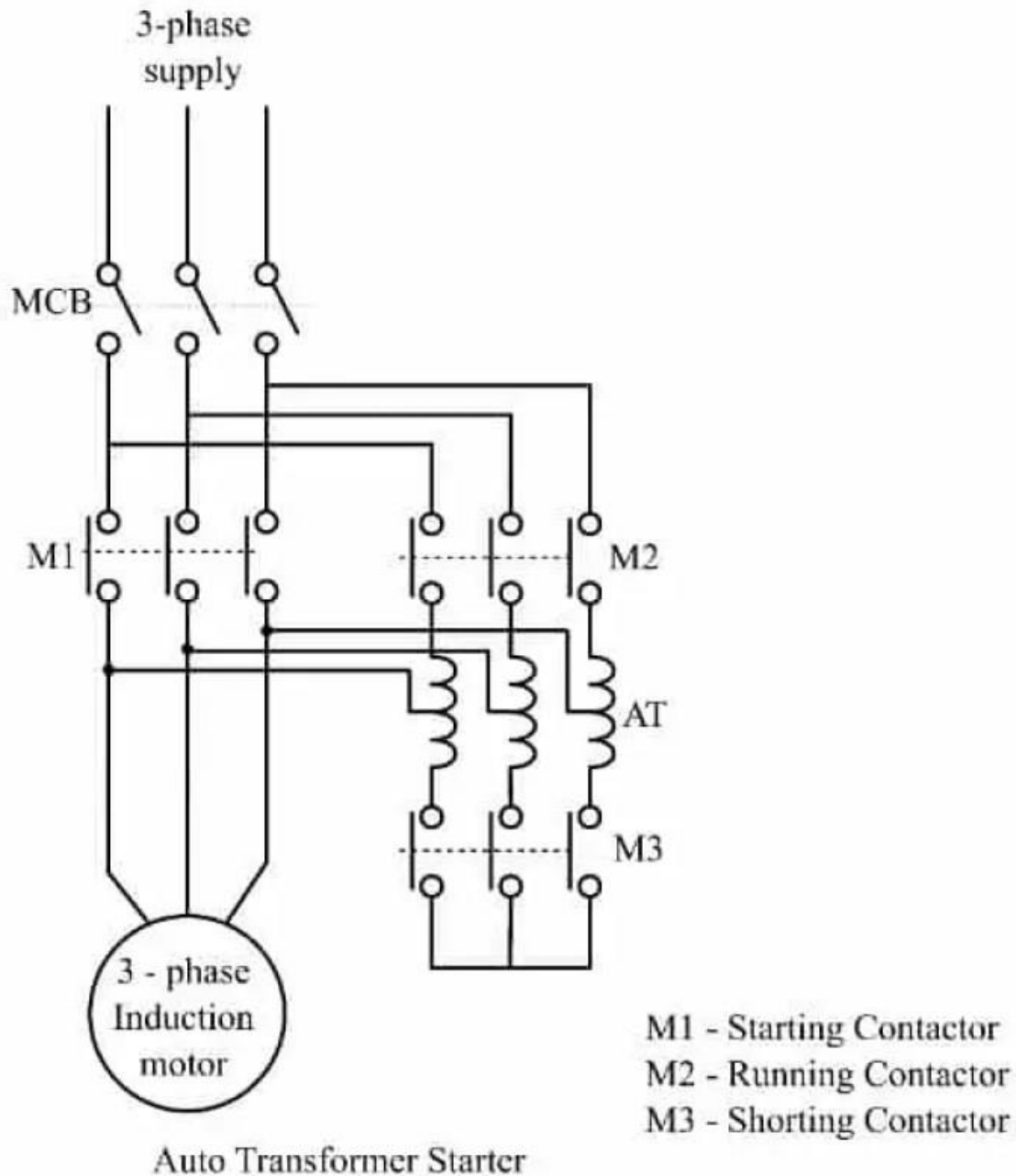


Star Delta Starter

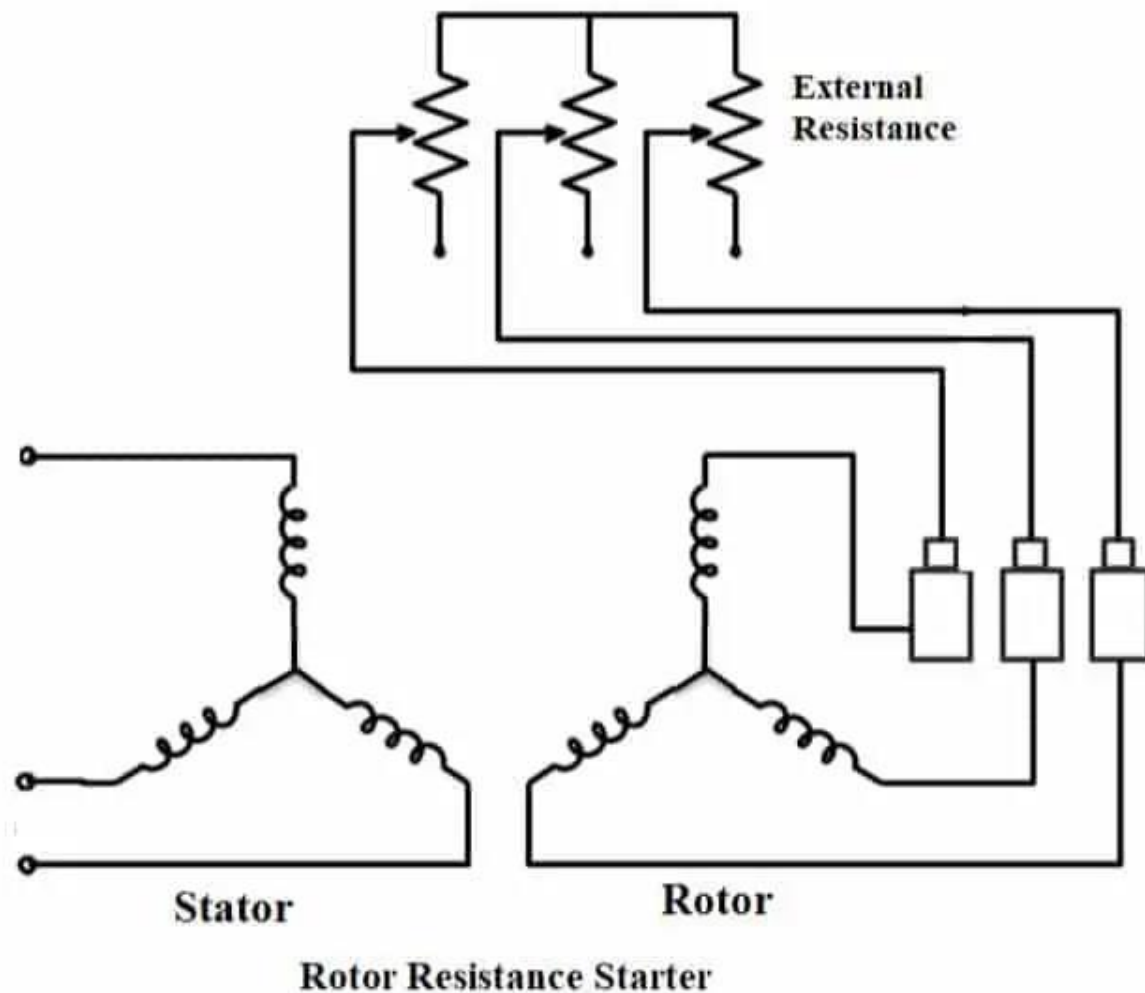
Star Delta starter This is the most common form of starter used for three phase induction motors.

It achieves an effective reduction of starting current by initially connecting the stator windings in star configuration which effectively places any two phases in series across the supply. Starting in star not only has the effect of reducing the motor's start current but also the starting torque. Once up to a particular running speed a double throw switch changes the winding arrangements from star to delta whereupon full running torque is achieved. Such an arrangement means that the ends of all stator windings must be brought to terminations outside the casing of the motor.

Auto-Transformer Starting This method of starting reduces the start current by reducing the voltage at start up. It can give lower start up currents than star-delta arrangements but with an associated loss of torque. It is not as commonly utilised as other starting methods but does have the advantage that only three connection conductors are required between starter and motor.



Rotor Resistance Starter If it is necessary to start a three phase induction motor on load then a wound rotor machine will normally be selected. Such a machine allows an external resistance to be connected to the rotor of the machine through slip rings and brushes. At start-up the rotor resistance is set at maximum but is reduced as speed increases until eventually it is reduced to zero and the machine runs as if it is a cage rotor machine.



REVERSAL OF DIRECTION OF ROTATION OF INDUCTION MOTOR

A three-phase induction motor takes a three-phase input electrical supply. Therefore, the three-phase induction motor is a self-starting motor because the three-phase electrical supply can produce a rotating magnetic field in the machine.

The direction of rotation of a 3 phase induction motor can be reversed by interchanging any two of the three motor supply lines. Such that the field rotates counterclockwise rather than clockwise. However, the number of poles and the speed at which the magnetic field rotates remain unchanged.

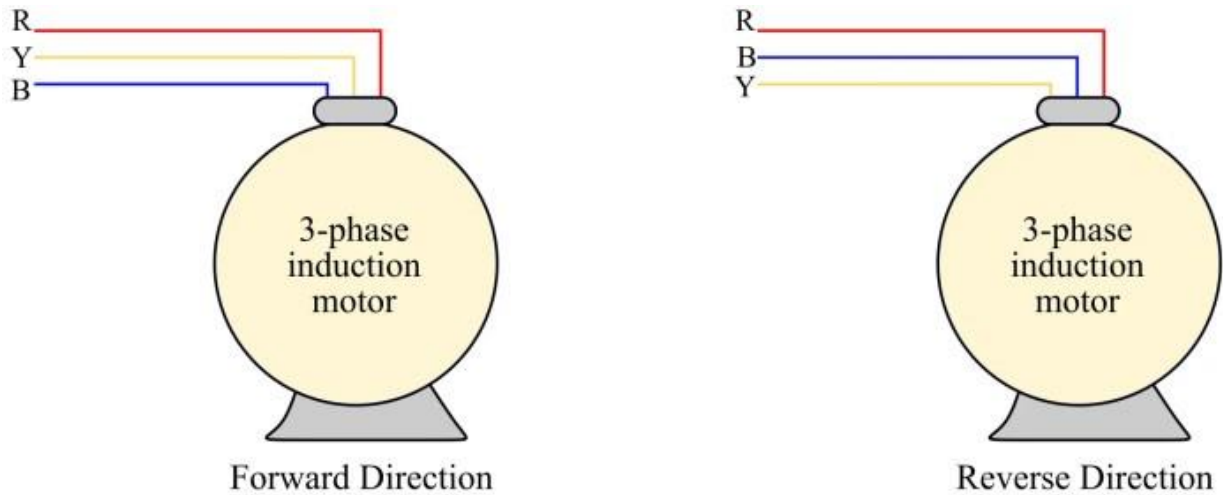


Figure 2 - Three-Phase Induction Motor

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Experiment: 10 Connection of Lamp by one way and Two Way Switching

Aim : To Study Connection of lamp by Two way switching. Switches, Wires, lamp, Switch Board.

Theory : _____.

TWO way switching means having or more switches in different locations to control one lamp. They are wired so that operation of either switch will control the light (s).

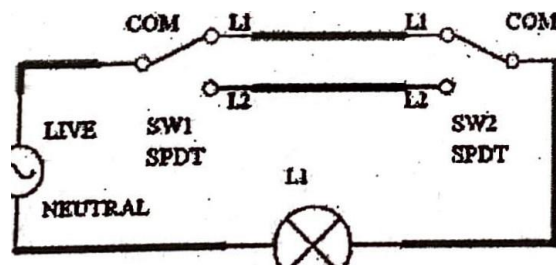
Two Way Switches :

A two light switch is a simple single pole "Changeover" switch with three terminals. These are typically labeled COM, L1 and L2 (some may label the L1 and L2 positions as I way and 2 way).

In one switch position the COM terminals is connected to L1. In the other switch position it changes Over so that COM is connected to L2. The design is a break before make type such that the connection to the new one is made.

Standard •two way switching circuit:

To achieve two way control the usual single switch is replaced with one of the following circuit a typical application is that of being able to switch the hall and landing lights on and off from both lights to be controlled from both locations could look like this.



Two way switches usually have terminal marked COM, L1 and L2. In the above diagram, the black core of the 3& has been connected to the COM terminal, and the brown to L1 and the grey to L2.

Some other ways: Three Switching Four Switching

Result: Learned and observed two way switching ,

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1. Fundamentals of Electrical engineering by Ashfaq Husain.
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